

Aaron Guo

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Education

Massachusetts Institute of Technology

B.S. / M.S. Computer Science (AI+D), B.S. Mathematics - GPA 5.0/5.0

Cambridge, MA

Sep 2023 - May 2027

- Relevant Coursework:
 - Design and Analysis of Algorithms, Computer Systems Engineering, Software Construction, Quantitative Methods of Natural Language Processing, Statistical Learning Theory, Computation Structures, Representation Inference and Reasoning in AI, Computational Cognitive Science, Fundamentals of Statistics

Work Experience

IMC Trading

Quantitative Trading Intern

Chicago, IL

Jun 2026 - Present

- Enhanced risk management strategy using quantitative and analytical techniques to minimize potential losses
- Modeled strategy performance by investigating market behavior and extracting predictive signals from trades data

MIT Mathematics

Undergraduate Researcher

Cambridge, MA

Sep 2025 - Aug 2026

- Optimized seasonal ARIMA models for US unemployment forecasting using AIC-based parameter selection and macroeconomic indicators (CPI and federal funds rates)
- Benchmarked deep learning, linear regression, and state-space models on SPY and Dow Jones market data
- Conducted comparative analysis of statistical and machine learning approaches for performance and interpretability in time series forecasting

Morgan Stanley

Software Engineering Intern

New York, NY

Jun 2025 - Aug 2025

- Calibrated design needs, developed, integrated, tested, and deployed low-latency, comprehensive applications used by business units and team members
- Applied variance reduction to validate Monte Carlo sample size reduction to build preparedness on a range of economic assumptions

MIT Media Lab (MAS)

Undergraduate Researcher

Cambridge, MA

Sep 2024 - Dec 2024

- Project SONAR (Decentralized AI)
- Developed components for a decentralized machine learning framework focused on collaborative model training
- Explored representation-sharing techniques to improve efficiency of split learning collaborative training algorithms

Kairos Health Capital Management

Analyst Intern

New York, NY

Jun 2024 - Dec 2024

- Designed and implemented algorithm to determine fair prices of monthly put options in the healthcare sector
- Identified arbitrage opportunities for using real-time option pricing
- Managed a portfolio of short put options across biotech ETFs with positive expected returns and low risk

Honors

- Mathematical Olympiad Program (MOP), USAJMO Winner (top 12), 5x USA(J)MO Qualifier
- HMMT February (Harvard and MIT Math Tournament): 5th place overall Individual, 1st place Team round, 1st place Guts Round, 1st place Sweepstakes
- Stanford Math Tournament (SMT): 1st place Geometry, 1st place Combinatorics, 3rd place Number Theory
- HPE CodeWars 2023 Virtual Contest Advanced Division: 1st place
- USAPhO Honorable Mention

Skills

- **Technical:** Python (NumPy, pandas, polars, matplotlib, sympy, pytorch), Java, C++, C, RISC-V assembly, SQL, R, JavaScript, Julia, Linux, bash, vim, cmd, git
- **Data Science / ML:** Deep Learning, Time Series Forecasting, Quantitative Trades Analysis, Statistical Modeling, Data Visualization, Monte Carlo Simulation, Reinforcement Learning, Natural Language Processing